

Wasi Omar Syed

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📍 <https://github.com/WasiOmar>

Education

Islamic University of Technology

Bachelor of Science in Software Engineering

CGPA 4.0 / 4.0 (4th semester)

Boardbazar, Gazipur

09/2024 - 09/2028

Relevant Coursework

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Structured Programming

Systems: Database Management Systems I & II

Math Foundations: Discrete Mathematics, Linear Algebra, Differential Calculus

Technical Skills

- **Programming:** Python, C++, Java, JavaScript
- **Machine Learning & AI:** PyTorch, Scikit-learn, YOLO, LangChain, LangGraph, CrewAI, FastMCP
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib, Jupyter Notebook
- **Backend & Deployment:** FastAPI, Node.js, Express.js, Next.js
- **Frontend & App Development:** React, Next.js, Tailwind CSS, Flutter
- **Databases:** MongoDB, PostgreSQL
- **DevOps & Cloud:** AWS (basic services), Docker, Kubernetes (fundamentals), Terraform (IaC)
- **Tools & Platforms:** Git, GitHub, n8n
- **Mathematical & Statistical Foundations:** Probability, Bayesian Inference, Hypothesis Testing, Markov Models, Statistical Inference

Projects

PawFect Match - A full-stack web application designed for pet adoption

Link: <https://deploy-wkrs.vercel.app/>

- Implemented AI-based matching system to recommend suitable pet-owner matches
- Developed “swipe-based” adoption flow (Tinder-style interaction model)
- Designed backend logic for pet attribute detection and user matching system
- Implemented real-time chat using web socket

Student-Driven Marketplace – Full-Stack AI-Powered Marketplace Platform

Link: <https://pawfectmatch-f1814.web.app>

- Built a full-stack marketplace platform for students to buy/sell items with authentication, listings, messaging, and admin moderation
- Developed frontend using React, Next.js, Tailwind CSS with responsive UI and protected routes
- Built backend using Node.js, Express.js, MongoDB with JWT authentication, role-based access control, and REST APIs
- Integrated real-time features using Socket.IO for messaging and notifications
- Implemented AI assistant and smart search features to enhance user discovery experience
- Designed scalable architecture with file uploads (Cloudinary/ImgBB), and security middleware (CORS)
- Deployed production-ready system on Vercel with CI-style deployment workflow

Deep Learning Lab – CNNs, Dense Networks & Transformers

GitHub: github.com/WasiOmar/Deep-Learning-Lab

- Implemented Transformer from scratch (Encoder + Decoder + Positional Embedding)
- Built and trained Convolutional Neural Networks (CNNs) for image classification using CIFAR-10 dataset
- Compared traditional Neural Networks vs CNNs for image recognition performance (Mini Project: Image Classification)
- Implemented Dense Neural Network-based image classifier and evaluated on benchmark dataset
- Applied Transformer-based architecture for NLP task (IMDB sentiment analysis)
- Gained practical understanding of deep learning architectures across vision and NLP domains

Hands-on Machine Learning (from scratch implementations)

GitHub: github.com/WasiOmar/hands-on-ml

- Implemented core ML algorithms from scratch using Python & NumPy (Linear/Logistic Regression, Gradient Descent, PCA, Decision Trees, etc.)
- Recreated key models without libraries to deeply understand mathematical foundations and optimization techniques
- Applied ML techniques on real datasets (Titanic, California Housing, Customer Churn, etc.)
- Compared classical ML models (e.g., Decision Trees vs Random Forest, Lasso vs Ridge)
- Built full ML pipeline notebooks including EDA, preprocessing, training, and evaluation

FeatherDB | C++17 | Custom Relational Database Engine

Role: Parser & Query Engine Lead | <https://github.com/SaifSarker11/FeatherDB>

- Designed and implemented the full SQL parsing pipeline from scratch: tokenizer → AST → query executor – no external libraries
- Built a recursive query executor supporting nested subqueries in both FROM and WHERE clauses
- Implemented in-memory ORDER BY sorting via Quicksort
- Enforced relational integrity at execution time: PRIMARY KEY, UNIQUE, and FOREIGN KEY constraints with RESTRICT delete policy

Certifications

- Harvardx CS50AI
 - Completed 12 AI projects using Python
 - Covered search algorithms, knowledge representation, optimization, machine learning, neural networks, and NLP
- Data Structures & Algorithms in C++ –Apna College (2025)
Problem Solving & Competitive Programming
- YouKnowWho Academy– Completed comprehensive training in algorithms, data structures, and problem-solving techniques (2025).

July 2025

Achievement

- **IUTCS CodeSprint 2025 – Hackathon (Runners-up)**

Project: PawFect Match – AI-powered pet adoption platform

Served as Project Lead, overseeing system design and implementation